METHYLENE BLUE

Serotonin syndrome with SSRIs

Blue-green urine

CHEMICAL NAME 3,7-bis(dimethylamino)phenothiazin-5-ium chloride (methylthioninium chloride)

CLASS Antidote (reducing agent); diagnostic dye

TRADE NAMES Proveblue · ProvayBlue · Methylene blue 1% injection

THERAPEUTIC USES / INDICATIONS

- Acquired methaemoglobinaemia (drug/chemical-induced; nitrites, dapsone, local anaesthetics, nitrates) with MetHb >20–30% or symptoms.
- Ifosfamide-induced encephalopathy.
- Refractory vasoplegic shock (off-label).
- Surgical dye (sentinel node, fistula/ureter identification).

ADMINISTRATION

- **IV** slowly over 5 min (diluted in glucose 5% — not saline), via large vein.
- Repeat after 1 h if needed.

DOSAGES

- **Methaemoglobinaemia:** 1–2 mg/kg IV over 5 min; repeat once after 1 h if needed; max 7 mg/kg total.
- **Infants <3 months:** 0.3–0.5 mg/kg.
- **Ifosfamide encephalopathy:** 50 mg IV every 4–6 h.

PHARMACOKINETICS

- **Onset:** MetHb falls within 30–60 min.
- Reduced in tissues to leucomethylene blue (active).
- Excreted in urine and bile (blue-green urine and stools). **t½:** 5–24 h.

PHARMACODYNAMICS

- Reduced by **NADPH-methaemoglobin reductase** to leucomethylene blue, which reduces methaemoglobin (Fe^{3+}) back to haemoglobin (Fe^{2+}) → restores oxygen carrying.
- Potent MAO-A inhibitor (serotonin syndrome); inhibits nitric oxide/guanylate cyclase (vasoplegia).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Blue-green urine, blue skin/mucosae.
- Nausea, abdominal pain, dizziness.
- Pain at injection site.

ADVERSE EFFECTS (SERIOUS)

- Serotonin syndrome with serotonergic drugs (SSRIs, SNRIs, clomipramine).
- Haemolysis — and ineffectiveness; in G6PD deficiency.
- Paradoxical methaemoglobinaemia (high doses).
- Interferes with pulse oximetry (falsely low SpO_2).

CONTRAINDICATIONS & CAUTIONS

- G6PD deficiency (haemolysis; ineffective).
- Methaemoglobinaemia due to nitrite in cyanide treatment.
- Hypersensitivity.
- Use with serotonergic drugs (relative — weigh risk).

KEY INTERACTIONS

- **SSRIs, SNRIs, TCAs, MAOIs, linezolid, tramadol, pethidine:** serotonin syndrome.

NURSING MANAGEMENT

- Check G6PD status if possible; review serotonergic drugs.
- Monitor MetHb level (co-oximetry), ABG; pulse oximetry unreliable.
- Monitor for serotonin toxicity (agitation, hyperthermia, clonus).
- Warn about urine colour.

PATIENT EDUCATION

- Urine and stools may turn blue-green for a day or two — this is harmless.
- Report agitation, confusion, fever or muscle twitching.

KEY POINTS

- Reduces MetHb to Hb.
- Contraindicated in G6PD deficiency.
- 1–2 mg/kg IV over 5 min.
- MAO inhibitor; serotonin syndrome; blue urine.

SOURCES CHECKED **[1001]** Methylthioninium chloride Proveblue 5 mg/ml solution for injection (UK SmPC) **[1002]** Methylene blue, IV (US label) **[1003]** Methylene blue, IV (US label, Amneal Pharmaceuticals LLC)

Full references p. 432–457